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A Machine Learning-Based Diagnosis of Leprosy: Study

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***Abstract*— Leprosy is a neglected skin infectious disease caused mainly by Mycobacterium leprae. It is annually reported in the South-East Asia Region in the majority and underserved areas. Early diagnosis is essential to eliminate leprosy since the real causes of contamination besides the untreated patients are unclear and it can cause a lot of damage if it is not treated. Considering a diagnostic done from images could be appropriate because the current diagnosis is clinical and sometimes in case of complication. AI/ML has proven to be reliable in predictions or diagnostics of human infections from numerical data making use of algorithms that can analyze, identify patterns, and make predictions with minimal human intervention. Among those algorithms, the Convolutional Neural Network and the Support vector machine have been chosen for the leprosy image analysis in this paper due to their performance in analyzing images.**

Keywords*— *Leprosy diagnosis, Machine Learning, Image-based diagnostic, imaging techniques.

I. INTRODUCTION

Leprosy, also known as Hansen's disease, is a neglected tropical disease (NTD) that continues to pose a public health challenge worldwide. Despite global eradication targets set by the World Health Assembly resolution 44.9 to eliminate leprosy as a public health problem by the year 2000 and in most countries by 2010 [1], the disease still affects over 200,000 people annually across more than 120 countries, with South Asia alone accounting for nearly half of all new cases as of 2023 [2]. The persistence of leprosy is often attributed to delayed diagnosis, insufficient public awareness, and limited healthcare access, especially in socioeconomically disadvantaged or geographically isolated populations. According to some statistics, Wenshan (China) and Nepal have shown that the disease disproportionately affects ethnic minorities like the Zhuang and Miao, who often lack adequate medical services and education [3].

Although multidrug therapy (MDT) has been the standard treatment and has proven effective in many cases, it has not been sufficient to eliminate the disease. The root causes and risk factors for continued transmission remain partially understood, which hampers prevention efforts. Compounding the issue is the scarcity of trained dermatologists in many low- and middle-income countries, making early diagnosis difficult [1,3,4]. Leprosy diagnosis is typically based on the detection of one or more of the three cardinal signs defined by the World Health Organization (WHO), including loss of sensation in a skin lesion, thickening of peripheral nerves, and the presence of acid-fast bacilli in skin smears. While laboratory and clinical evaluations are still essential, certain diagnostic signs such as nerve thickening require imaging techniques, since they cannot be identified through blood tests.

In recent years, artificial intelligence (AI), particularly machine learning (ML), has shown promise in supporting medical diagnostics, including leprosy. ML is a subset of AI that involves algorithms capable of learning patterns from data to make predictions or decisions without being explicitly programmed [5,6]. A variety of data formats can be used in ML models, including structured records and medical images. This paper focuses specifically on image-based applications of ML in leprosy diagnosis.

In medical imaging, an image can be represented as a two-dimensional function $F(x, y)$, where (x, y) are spatial coordinates, and the function's value at any point reflects the intensity at that location. Processing such images typically involves several stages: acquisition, preprocessing, segmentation, feature extraction, and classification or prediction [7]. Various segmentation techniques have been successfully applied for medical imaging analysis. Himabindu et al. [8] proposed a threshold and segmentation-based X-ray imaging analysis for COVID-19 detection, demonstrating the role of feature extraction in classification accuracy (Acc). Similarly, studies like those of Himabindu and Sekar [9,10] have shown that cyst detection using ultrasound images can be efficiently achieved using active contour segmentation, which is relevant for lesion boundary detection in leprosy.

Since the pioneering work by Harjatin Singh Baweja and Tanvir Parhar in 2016, which employed convolutional neural networks (CNNs) for lesion classification [11], additional studies have explored various deep learning techniques for improved diagnostic Acc. in leprosy, 6 different types according to **W. H. Jopling** [12,13]. This paper reviews the current landscape of AI/ML applications in leprosy diagnosis, highlighting the algorithms used, imaging modalities applied**.**

II. RELATED WORKS

This study conducted a review to showcase the various Machine learning algorithms used for Leprosy detection. The goal of the study is to collect all the existing papers on AI/ML for leprosy diagnosis in the African and Asian populations between **2011 and 2024** to ensure relevance to current advancements in medical image processing and AI-based diagnostics. The analysis covers datasets, preprocessing techniques, feature extraction, and model comparisons. The review focused on studies published primarily on the population of South India, Brazil, and Africa.

Both experimental studies and applied research were included to provide a balanced understanding of the current capabilities and limitations of machine learning in leprosy diagnosis.

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A. Image Datasets Used

Many of the studies relied on clinical images of leprosy skin lesions, collected from hospitals, dermatology clinics, or open-source repositories. Baweja and Parhar used a dataset from DermNetNZ.org that contained balanced images of leprosy and non leprosy lesions. The model he proposed was able to distinguish between skin disorders from similar-looking skin disorders such as ringworm, syphilis, or dermatitis. In this paper, the authors applied filters without giving detailed information about them. However, their goal of applying the filters was to stay focused on various visual pat terns such as edges, curves, or strokes, favoring the building of a meaningful feature map across the RGB color channels. This step was essential to detect lesion patterns efficiently, wherever it is located in the pictures, to achieve an average Acc. of 91.6%, whereas the average Acc. for architecture with a lesser number of filters is 69.4% [14]. Barbieri and al. used high-resolution data from 222 patients (1229 images), including their demographic information and skin lesion photos. That dataset, for privacy reasons, access is limited and requires registration and ethical permission. Their processing methods include grouping rare categories with more common ones to handle incomplete metadata. Harmonizing the variables helped the author ensure that statistical models were not biased by very few cases from different backgrounds. Furthermore, excluded whenever necessary for particular images were lesion temperature or diameter, so guarantee algorithm input data consistency. They achieved an Acc. of 81.3%, 80.8%, and 77.9%, respectively, for Logistic Regression (LR), XGB, and Random Forest (RF) without applying any filters. To show the stability of the model, they ran the models multiple times and calculated a standard deviation, which gave 0.058,0.06,0.088, respectively. Others metrics such as AUC, sensitivity, and Specificity have been used for this work [15].

The study done by R. Beesetty, S. A. Reddy focused on a small clinical dataset using a few-shot learning approach, acknowledging the challenge of data scarcity in low-resource settings. In this study, the images were collected through a collaborative observational study with the Sivananda Rehabilitation Home (SRH) in Hyderabad, India, where researchers documented existing clinical practices without interfering with standard procedures. Patients were enrolled from multiple districts across Telangana through SRH-organized leprosy camps. The initial aim was to collect images from 1,000 participants (500 with confirmed leprosy and 500 with leprosy-like conditions), but data collection was interrupted by the COVID-19 pandemic, resulting in a limited dataset. The dataset included diverse forms of leprosy, such as tuberculoid, border line, and lepromatous types. Only patients untreated or treated for less than three months were included to ensure the lesions reflected active disease. Ethical approval was obtained from the Ramdev Rao Hospital's Institutional Ethics Committee, and informed consent was secured from all participants (or guardians for minors). Instead of applying filters manually, they used high-resolution cameras with a polarizing filter in two mutually perpendicular directions for each image. The non-application of filters led to an Acc. of 26.56% of the test set using Baseline1 (nearest neighbor) despite the use of the ADAM optimizer. In their work "Heuristic Segmentation Assisted Deep-Spatial Feature Learning Model for Leprosy Detection," Jitendra and al. (2024) used a clinically acquired dataset of 220 leprosy lesion images. RGB pictures of skin lesions that have been identified as either leprosy or normal skin make up the dataset. Although precise numbers and demographics are not specified in the paper, the images were taken from dermatological settings. They make use of data augmentation techniques to remedy the dataset deficit and enhance the model's generalization. They used flipping, rotation, and scaling of images to create variability in the training set. This method helps simulate different lesion orientations and positions, which may occur naturally in clinical photographs. Since the methodology focuses on ROI segmentation using a Fuzzy C-Means (FCM) clustering algorithm driven by the Firefly heuristic, each image included a region of interest (ROI). The paper doesn't specify whether multiple clinicians confirmed the presence or type of the lesion or whether the dataset is publicly available, but the segmented ROIs were then used for feature extraction and classification. Banerjee and al., for their part, in the paper Skin Diseases Detection using LBP and WLD- And Ensemble Approach, collected their dataset images from the out door patients of the School of Medicine (STDM), Kolkata, over one year from April 2011 to March 2012. In their work, basic normalization techniques were performed. Banerjee and al. standardized image dimensions by resizing and cropping the lesion areas to 144*144 pixels. Then, the images were converted to grayscale to simplify the processing and reduce computational load. The Discrete Cosine Transform (DCT) was one of the methods used to extract features for images. The DCT serves as a tool of analyzing images by separating the image into distinct frequency bands or spectral sub-bands. The DCT is used to extract meaningful features from the images. At first, the DCT is applied to the entire image, and then some of the coefficients are used to create the feature vector. For instance, for an M*N-dimensional image, the DCT is computed as:

$$F(a,b) = \frac{1}{\sqrt{MN}} \propto (a) \propto (b) \sum_{x=0}^{M-1} \sum_{y=0}^{N-1} Q$$

$$\text{And } Q = f(x,y) \times \cos\left(\frac{(2x+1)u\pi}{2M}\right) \times \left(\frac{(2y+1)v\pi}{2N}\right)$$

Where a=0, 1, ..., M and b=0, 1, …, N and $\propto$ (J) is defined by:

$$\propto (J) = \begin{cases} \frac{1}{\sqrt{2}}, & J = 0 \\ 1, otherwise \end{cases}$$

B. Comparison Between Models

Different classical ML models have been used to identify and classify leprosy and its subtypes. Among these algorithms, the Support Vector Machines (SVM) and Logistic Regression (LR) emerged as the most used and effective performers(Table 1). According to Jalpa Metha's study, SVM achieved an Acc. of 97.80%, while LR and ANN reached 95.80% and 95.56% respectively, showing their strength in handling feature-rich data derived from WHO and Jopling classification systems. Jitendra and al. demonstrated the effectiveness of a texture-based model by using Random

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Forest combined with GLCM (Gray-Level Co-occurrence Matrix) features. In this study, the classification achieved an Acc. of 92.7% and a high F1 score of 0.94. This showcases the value of handcrafted texture features for medical image classification, especially when large datasets are unavailable. In contrast, the study by Beestty using unspecified baseline models on a limited clinical dataset reported a lower Acc. of 73.13%. This reflected the challenges posed by data scarcity and the limited richness of input features, which hindered model performance. Banerjee and al. applied a multi-class SVM to grayscale images processed with LBP and WLD texture descriptors. The model achieved an Acc. of 83.97%, suggesting that ensemble approaches using multiple handcrafted features can deliver reliable performance even without color information. In studies that used structured metadata rather than images [figure 1], Barbieri and al. reported an AUC of 88.6% using Logistic Regression, highlighting that clinical variables like age, lesion location, and symptoms can provide strong diagnostic signals when image data is limited or unavailable. Despite the promising results, several **recurring limitations** were identified:

- **Small and imbalanced datasets** were a frequent constraint, reducing the robustness and generalizability of models.
- **Lack of standardization** in feature extraction methods, preprocessing steps, and evaluation protocols made cross-study comparison difficult.
- **Insufficient reporting** of key performance indicators such as sensitivity, specificity, and confusion matrices in some studies limited the depth of performance analysis.
- **Restricted dataset access** and lack of open-source availability hindered replication and benchmarking of results.

In summary, models that leveraged well-prepared metadata or carefully extracted texture features performed consistently better. However, further improvements depend on addressing dataset limitations, adopting consistent evaluation criteria, and increasing transparency in model design and data preparation.

III. Discussion

The application of machine learning (ML) techniques in leprosy diagnosis has shown promising results when a dataset is available. Classical algorithms such as SVM, Logistic Regression, and Random Forest have proven to be effective when paired with texture features or clinical metadata. Their ability to provide interpretable results and operate on low-resource datasets makes them especially valuable in the context of leprosy, a neglected tropical disease that often affects underserved populations. One of the key strengths of current ML-based approaches lies in their adaptability. Even in the absence of large datasets, models utilizing handcrafted features like GLCM, LBP, or structured demographic inputs have achieved respectable Acc. and AUC scores. Moreover, these models are computationally efficient and suitable for deployment in resource-limited settings. However, despite these advantages, several weaknesses persist. The most pressing issue is the limited size and diversity of datasets used in most studies. With few publicly available image repositories and significant data imbalance, many models suffer from overfitting and cannot generalize across broader populations.

Cultural, ethnic, and geographical variations in skin conditions are rarely addressed, which limits model performance outside the original sample group. Another weakness is the limited interpretability of some ML models. While classical algorithms are more transparent than deep learning models, the decision-making process, especially in ensemble models, can still be difficult for clinicians to understand or trust without clear explanations. In terms of gaps in the literature, very few studies have focused exclusively on leprosy using classical ML techniques. Much of the existing work either combines it with broader skin disease classification or uses deep learning without comparing results against traditional methods. Additionally, the lack of standardized datasets, both in format and clinical labeling, prevents consistent benchmarking. Lastly, no study to date has reported a large-scale clinical validation, which is essential for real-world adoption and regulatory approval.

A. Machine Learning Models Applied

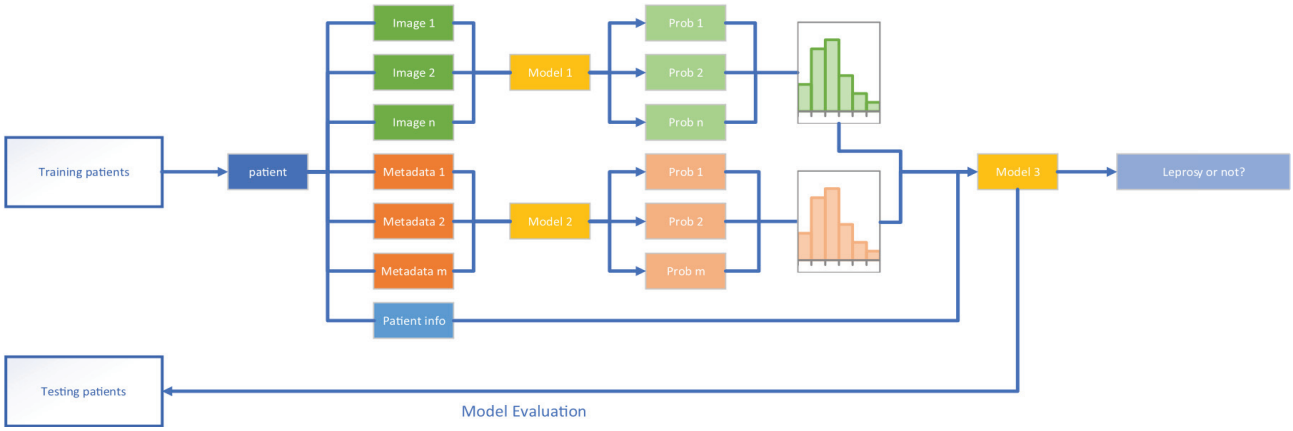


Fig. 1. Comprehensive data modeling summary encompassing the complete dataset of 22 patient records. Images (model 1) or metadata (model 2) from 182 patients were used to train the algorithms in a training dataset, while 40 patients were kept as a testing group for validation in model 3. The outputs (histogram) of models 1 and 2 were used to evaluate the Acc. and AUC [15].

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TABLE I. SUMMARY OF MACHINE LEARNING MODELS APPLIED WITH THEIR RESULTS ALONG WITH DATA SET USED FOR LEPROSY DIAGNOSIS FROM SKIN LESION IMAGES

Study	Algorithm Used	Dataset (name of the dataset)	Evaluation Metrics (%)	Key Results	Limitations
Baweja and Parhar [14]	CNN	Skin lesion images from dermnetnz.org and web scraping of Google images.	Acc. 91.6	The models with more filters perform well than others	The authors didn't provide enough evaluation metrics
Barbieri et al. [15]	LR, XGBoost, RF	Collected from the Brazilian leprosy national referral center, 222 patients(1229 images)	Acc.: LR: 81.3 XGB:80.8 RF:77.9	Metadata-only models performed well; LR gave the highest Acc.	The restriction of the dataset, along with limited preprocessing transparency and model integration were part of the limitations.
Jalpa Metha[16]	SVM, LR, ANN	Dataset name not provided.	Acc.: LR: 95.80 SVM: 97.80 ANN: 95.56	The SVM performed well	Limited dataset
R. Jitendra et al.[17]	ResNet101 (Hybrid model), Random Forest for Classification	Clinical skin lesion images containing confirmed leprosy.	RF: Acc.: 97.20, Precision: 99.69, Recall: 95.82, F1: 97.80	The hybrid feature framework combined GLCM texture descriptors and deep features yield robust representations. RF classifier used as shown effectiveness in working on images and class-imbalance issues	The study reports only Acc. and F-Measure. Other critical metrics (e.g., sensitivity, specificity, confusion matrices) are not provided, limiting a full clinical evaluation and also the dataset is not publicly accessible for future studies
R. Jitendra et al. [18]	Random Forest (GLCM features)	220 lesion images from hospitals/online; texture descriptors via GLCM	Acc.: 92.7 Precision: 89.5 Recall: 92.1 F1: 94	Strong texture-based classification using handcrafted features	Some of the limitations were the limited dataset and the lack of demographic diversity.
R. Beesetty[19]	Baseline classical models (Nearest Neighbor)	Images were gathered during a collaborative observational medical study with Sivananda Rehabilitation Home in Hyderabad, India.	Acc.: 73.13%	Baseline models underperformed due to data scarcity	The dataset has a low sample size
Banerjee et al.[20]	Multi-class SVM	STM, Kolkata; images resized (144×144), grayscale; texture features (LBP, WLD)	83.97 of Acc.	The ensemble texture-based approach showed effectiveness	The authors did not provide enough evaluation metrics.

IV. FUTURE DIRECTIONS

To advance the role of ML in leprosy diagnosis, several research avenues should be explored:

1. Development of Annotated and Diverse Datasets: One of the most urgent needs is the creation of large, well-labeled, and openly accessible datasets for leprosy image. These should include diverse patient demographics, lesion types, and acquisition settings to improve model training and generalizability.
2. Hybrid Diagnostic Models: Future systems could combine classical machine learning with rule-based expert systems to improve interpretability and reliability. For example, features identified by dermatologists could be integrated with algorithm outputs to build trust among clinical users.
3. Multimodal Data Integration: Incorporating patient metadata such as age, gender, geographic origin, and clinical history alongside image data could significantly improve prediction Acc. and risk stratification. This is particularly important for neglected diseases with varied symptom presentation.
4. Model Explainability and Clinical Integration: Future research should also focus on making ML decisions interpretable. Visualization tools, confidence scores, and decision trees could help clinicians understand why a model makes a particular diagnosis, thus enhancing its integration into clinical workflows.

V. CONCLUSION

This review explored the use of classical machine learning techniques for the diagnosis of leprosy using image-based and structured clinical data. The reviewed studies demonstrated that traditional algorithms such as Support Vector Machines, Logistic Regression, and Random Forest can achieve commendable diagnostic performance when applied to well-prepared datasets. Particularly, models leveraging texture descriptors like GLCM and LBP, or relying on structured patient metadata, were shown to be effective in classifying lesion types and supporting clinical decision making. Despite the growing interest in deep learning, this review highlights the continued relevance and practicality of non-deep learning methods, especially in contexts where data is scarce or computational resources are limited. These models are not only faster and more interpretable but also suitable for early-stage screening tools in regions where healthcare access is limited. Moving forward, future research should prioritize the development of standardized, annotated datasets, promote the integra tion of multimodal data sources, and ensure models are evaluated through clinical trials or field validations. Classical machine learning, when applied thoughtfully, still holds significant promise in enhancing early detection and reducing diagnostic delays for neglected diseases like leprosy.

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